

COUNTEREXAMPLES TO FIVE CONJECTURES OF GRAFFITI.PC ON DOMINATION

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ABSTRACT. Written on the Wall II, the list of conjectures of the program Graffiti.pc, records Conjectures 319, 352, 358, 359 and 427 as open. We show that all five are false. The counterexamples are caterpillars for Conjectures 352, 358, 359 and 427, and graphs built from complete graphs for Conjecture 319; each comes in an infinite family, and for Conjectures 358, 359 and 427 the difference between the two sides is unbounded.

1. INTRODUCTION

Graffiti.pc is a program written by DeLaViña in 2001 [4] after Fajtlowicz's Graffiti [9]; it proposes inequalities between graph invariants that hold on its database of graphs. Its conjectures are collected, with their status, in DeLaViña's list *Written on the Wall II* [5]. Several hundred of them concern total domination, and some have been settled [7, 6, 8]. This note disproves five conjectures that the list still marks as open (in its version of 26 July 2026): four on total domination and one on independent domination. The proofs are by hand. As in [2], every finite statement is also re-checked by computer and in the Lean proof assistant; the computations are described in Section 7 and Appendix A.

All graphs are finite, simple and connected. For a vertex v of G , $N(v)$ is its set of neighbours and $N[v] = N(v) \cup \{v\}$; for $X \subseteq V(G)$, $N(X) = \bigcup_{v \in X} N(v)$ and $\langle X \rangle$ is the subgraph induced by X . A set D is *dominating* if every vertex outside D has a neighbour in D , and *totally dominating* if every vertex of G has a neighbour in D . The domination number $\gamma(G)$ and the total domination number $\gamma_t(G)$ are the least sizes of such sets [3, 10]; the independent domination number $i(G)$ is the least size of a dominating set that is independent. A total dominating set is *minimal* if no proper subset is a total dominating set; since every superset of a total dominating set is one, this holds as soon as no single vertex can be removed. G is *well totally dominated* if all its minimal total dominating sets have the same size [1]. A *pendant* vertex or *leaf* has degree one, and a *support vertex* is a neighbour of a leaf. We write $d(u, v)$ for the distance, $\text{ecc}(v) = \max_u d(v, u)$ for the eccentricity, and $C(G)$ for the *center*, the set of vertices of least eccentricity. Following [5], the eccentricity of a set X is

$$\text{ecc}(X) = \max_{v \notin X} \min_{x \in X} d(v, x).$$

The number of vertices at even distance from v , including v itself, is denoted $\text{de}(v)$; the list calls it $\text{dist}_{\text{even}}(v)$. In a tree T , $L(T)$ is the set of leaves, $S(T)$ the set of support vertices, $D_2(T)$ the set of vertices of degree two, and $M(T)$ the set of vertices of maximum degree. Finally $\text{iso}(H)$ and $\text{c}(H)$ are the numbers of isolated vertices and of components of a graph H .

The five conjectures read as follows in [5].

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Conjecture 352. If T is a tree on $n > 2$ vertices, then $\gamma_t(T) \geq c\langle N(D_2) \cup D_2 \rangle + \lceil \frac{1}{2} \text{ecc}_{\text{avg}}(M) \rceil$, where $\text{ecc}_{\text{avg}}(M)$ is the average eccentricity of the vertices of maximum degree.

Conjecture 358. If T is a tree on $n > 2$ vertices, then $\gamma_t(T) \geq \frac{1}{2} \text{ecc}(C) + \text{iso}\langle S \rangle$.

Conjecture 359. If T is a tree on $n > 2$ vertices, then $\gamma_t(T) \geq \frac{1}{2} \text{ecc}(C) + c\langle S \cup L \rangle$.

Conjecture 319. If G has $n > 1$ vertices and $\max_v \text{de}(v) = \gamma(G)$, then G is well totally dominated.

Conjecture 427. If G has $n > 3$ vertices, then $i(G) \leq |E(C, V - C)| + \lfloor \frac{2}{3} |E(V - N(P))| \rfloor$, where P is the set of pendant vertices, $E\langle X \rangle$ is the edge set of $\langle X \rangle$ and $E(C, V - C)$ the set of edges with exactly one end in C .

The printed statement of Conjecture 427 introduces the set M of maximum-degree vertices and then uses the symbols C and P without defining them. In the neighbouring entries of the list, C is the center (Conjectures 430a–c) and P the set of pendant vertices (Conjecture 425d); we read the statement this way, and we show in Theorem 6.2 that it fails also if C stands for M .

Theorem 1.1. *Conjectures 352, 358, 359, 319 and 427 are false. More precisely:*

- (1) *for $a \equiv 3$ and $r \equiv 0 \pmod{4}$ with $r \geq 4$, the caterpillar $T(a, r)$ on $a + r + 11$ vertices has $\gamma_t = \frac{a+r+7}{2}$, one less than the right side of Conjecture 352 (Theorem 3.2);*
- (2) *for odd $k \geq 3$, the caterpillar Q_k on $7k - 2$ vertices has $\gamma_t = 3k$, while the right sides of Conjectures 358 and 359 both equal $3k + \frac{k-1}{4}$ (Theorem 4.2);*
- (3) *for $k \geq 3$, the graph G_k on $3k + 1$ vertices satisfies $\text{de}(v) = \gamma(G_k) = k + 1$ for every vertex v , and has minimal total dominating sets of sizes $k + 1$ and $2k$ (Theorem 5.2);*
- (4) *for $m \geq 2$, the caterpillar H_m on $4m + 1$ vertices has $i = 2m$, while the right side of Conjecture 427 equals 2 (Theorem 6.2).*

The smallest counterexamples are $T(3, 4)$ with 18 vertices, Q_3 with 19, G_3 with 10, and an 8-vertex tree for Conjecture 427. An exhaustive search (Section 7) shows that $T(3, 4)$ and Q_3 are the only counterexamples of their orders and that there are none smaller. Likewise G_3 is the only counterexample to Conjecture 319 with at most 11 vertices, and the 8-vertex tree the only counterexample to Conjecture 427 with at most 8 vertices.

2. TWO TOOLS

Lemma 2.1. *Every total dominating set of a tree contains all its support vertices, and every dominating set meets $\{s\} \cup (N(s) \cap L)$ for each support vertex s .*

Proof. A leaf ℓ has the single neighbour s , so a total dominating set must contain s , and a dominating set must contain ℓ or s . $\square$

Lemma 2.2. *Let D be a total dominating set of a graph, R a set of vertices, and Q a set of vertices whose neighbours all lie in R . If every vertex of R has at most two neighbours in Q , then $|D \cap R| \geq |Q|/2$.*

Proof. Each $q \in Q$ has a neighbour in D , which lies in $D \cap R$. Counting the pairs (q, x) with $q \in Q$, $x \in D \cap R$ and qx an edge gives $|Q| \leq \sum_{x \in D \cap R} |N(x) \cap Q| \leq 2|D \cap R|$. $\square$

We also use Jordan's theorem [11]: the center of a tree consists of the middle vertex, or the two middle vertices, of any longest path.

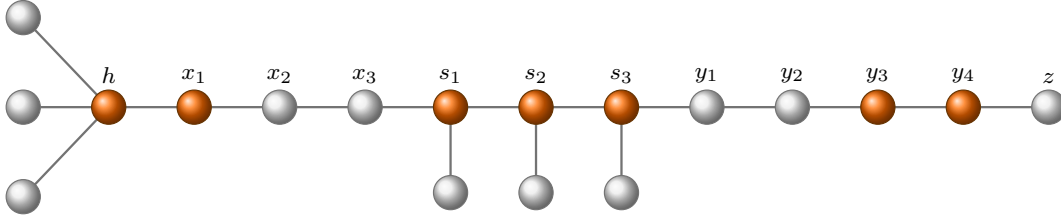


FIGURE 1. $T(3, 4)$, the smallest counterexample to Conjecture 352. The shaded vertices form a total dominating set of size 7; the right side of the conjecture is 8.

3. CONJECTURE 352

Definition 3.1. For $a, r \geq 1$ let $T(a, r)$ be the tree formed by a path

$$h, x_1, \dots, x_a, s_1, s_2, s_3, y_1, \dots, y_r, z$$

together with three leaves attached to h and one leaf ℓ_i attached to each s_i (Figure 1). It has $a + r + 11$ vertices.

Theorem 3.2. Let $a \equiv 3 \pmod{4}$ and $r \equiv 0 \pmod{4}$, $r \geq 4$. Then

$$\gamma_t(T(a, r)) = \frac{a + r + 7}{2} < \frac{a + r + 9}{2} = c\langle N(D_2) \cup D_2 \rangle + \left\lceil \frac{1}{2} \text{ecc}_{\text{avg}}(M) \right\rceil.$$

Proof. Write $T = T(a, r)$. The vertex h has degree 4, the vertices s_1, s_2, s_3 have degree 3, the x_i and y_j have degree 2, and all other vertices are leaves. Hence $M = \{h\}$, and $\text{ecc}(h) = d(h, z) = a + r + 4$, since every other vertex is closer to h . As $a + r$ is odd, $\lceil \frac{1}{2} \text{ecc}(h) \rceil = \frac{a+r+5}{2}$. Next, $D_2 = \{x_1, \dots, x_a, y_1, \dots, y_r\}$ and

$$N(D_2) \cup D_2 = \{h, x_1, \dots, x_a, s_1\} \cup \{s_3, y_1, \dots, y_r, z\}.$$

The two sets on the right induce paths, and no edge joins them because s_2 lies in neither and $s_1 s_3$ is not an edge. So $c\langle N(D_2) \cup D_2 \rangle = 2$, and the right side of Conjecture 352 equals $\frac{a+r+9}{2}$.

Upper bound. The set

$$D = \{h, x_1, s_1, s_2, s_3\} \cup \{x_{4j}, x_{4j+1} : 1 \leq j \leq \frac{a-3}{4}\} \cup \{y_{4j-1}, y_{4j} : 1 \leq j \leq \frac{r}{4}\}$$

has $5 + \frac{a-3}{2} + \frac{r}{2} = \frac{a+r+7}{2}$ elements. It is a total dominating set: h dominates its leaves and x_1 ; x_1 dominates h and x_2 ; for $j \geq 1$ the pair x_{4j}, x_{4j+1} dominates $x_{4j-1}, \dots, x_{4j+2}$, so that $x_1, \dots, x_{a-1}$ are all dominated; s_1 dominates x_a, ℓ_1 and s_2 ; s_2 dominates s_1, s_3 and ℓ_2 ; s_3 dominates s_2, ℓ_3 and y_1 ; and the pair y_{4j-1}, y_{4j} dominates $y_{4j-2}, \dots, y_{4j+1}$, where y_{r+1} means z .

Lower bound. Let D be any total dominating set. By Lemma 2.1, $h, s_1, s_2, s_3 \in D$. Apply Lemma 2.2 to $R_1 = N(h) \cup \{x_2, \dots, x_a\}$ and $Q_1 = \{h, x_2, \dots, x_{a-1}\}$: the neighbours of h and of each x_i with $2 \leq i \leq a-1$ lie in R_1 , and a vertex of R_1 is adjacent to at most two vertices of Q_1 . Hence $|D \cap R_1| \geq \frac{a-1}{2}$. Likewise, with $R_2 = \{y_1, \dots, y_r, z\}$ and $Q_2 = \{y_2, \dots, y_r, z\}$, $|D \cap R_2| \geq \frac{r}{2}$. The sets $\{h, s_1, s_2, s_3\}$, R_1 and R_2 are disjoint, so $|D| \geq 4 + \frac{a-1}{2} + \frac{r}{2} = \frac{a+r+7}{2}$. $\square$

4. CONJECTURES 358 AND 359

Definition 4.1. For $k \geq 1$ let Q_k be the caterpillar whose spine is the path

$$s_1, c_1, s'_1, u_1, w_1, s_2, c_2, s'_2, u_2, w_2, \dots, u_{k-1}, w_{k-1}, s_k, c_k, s'_k$$

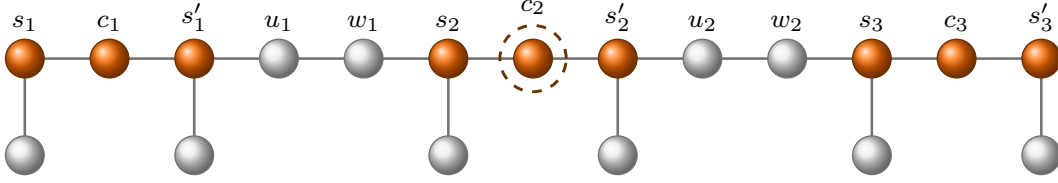


FIGURE 2. Q_3 , the smallest counterexample to Conjectures 358 and 359. The center c_2 is circled; the shaded vertices form a total dominating set of size 9, while both conjectures demand $\frac{19}{2}$.

with one leaf attached to each s_j and to each s'_j (Figure 2). It has $7k - 2$ vertices. We call $\{s_j, c_j, s'_j\}$ the j -th block.

Theorem 4.2. *Let $k \geq 3$ be odd. Then $\gamma_t(Q_k) = 3k$, $\text{ecc}(C) = \frac{5k-1}{2}$ and $\text{iso}\langle S \rangle = \text{c}\langle S \cup L \rangle = 2k$. Hence the right side of both Conjecture 358 and Conjecture 359 equals $3k + \frac{k-1}{4} > \gamma_t(Q_k)$.*

Proof. The leaves are the $2k$ attached vertices, so $S = \{s_j, s'_j : 1 \leq j \leq k\}$. Two support vertices are at distance at least 2 (they are separated by c_j or by u_j, w_j), so $\langle S \rangle$ has no edges and $\text{iso}\langle S \rangle = 2k$; in $\langle S \cup L \rangle$ each support vertex is joined only to its own leaf, so $\text{c}\langle S \cup L \rangle = 2k$.

The spine has $2k + 3(k-1) = 5k - 3$ edges, and every vertex lies on the spine or is a leaf attached to it; the spine ends s_1 and s'_k carry leaves. So the path from the leaf at s_1 to the leaf at s'_k , of length $5k - 1$, is a longest path. Since k is odd, $5k - 1$ is even, and by Jordan's theorem C is the single vertex at distance $\frac{5k-1}{2}$ from both ends; this is c_m with $m = \frac{k+1}{2}$, because the leaf at s_1 is at distance $1 + 5(m-1) + 1 = \frac{5k-1}{2}$ from c_m . Thus $\text{ecc}(C) = \text{ecc}(c_m) = \frac{5k-1}{2}$.

The set B of all $3k$ block vertices is a total dominating set: each leaf is dominated by its support vertex, s_j and s'_j by c_j , c_j by s_j , u_j by s'_j , and w_j by s_{j+1} . Conversely, let D be a total dominating set. By Lemma 2.1, $S \subseteq D$. For each j the vertices s_j, s'_j need neighbours in D , and these lie in $X_j = N(s_j) \cup N(s'_j)$, which contains no support vertex. The sets $X_1, \dots, X_k$ are pairwise disjoint, because each vertex outside S is adjacent to support vertices of only one block: c_j to s_j, s'_j ; u_j to s'_j ; w_j to s_{j+1} ; a leaf to its support vertex. Hence $|D| \geq 2k + k = 3k$, and $\gamma_t(Q_k) = 3k$.

Finally $\frac{1}{2} \text{ecc}(C) + 2k = \frac{5k-1}{4} + 2k = 3k + \frac{k-1}{4}$. $\square$

Remark 4.3. For even $k \geq 4$ the center of Q_k is the edge $u_{k/2}w_{k/2}$, $\text{ecc}(C) = \frac{5k-2}{2}$, and the same argument gives right sides $3k + \frac{k-2}{4} > 3k = \gamma_t(Q_k)$. For $k = 2$ the two sides are equal.

5. CONJECTURE 319

Definition 5.1. For $k \geq 3$ let G_k consist of a complete graph on $c_1, \dots, c_k$, a vertex x , and for each i a path x, p_i, q_i, c_i (Figure 3). It has $3k + 1$ vertices.

Theorem 5.2. *For $k \geq 3$, every vertex v of G_k has $\text{de}(v) = k+1$, and $\gamma(G_k) = \gamma_t(G_k) = k+1$. The sets $\{x, p_1, c_2, \dots, c_k\}$ and $\{p_1, \dots, p_k, q_1, \dots, q_k\}$ are minimal total dominating sets of sizes $k+1$ and $2k$. Hence G_k satisfies the hypothesis of Conjecture 319 and is not well totally dominated.*

Proof. Distances. Let $i \neq j$. The following lists, read off from the paths through x and through the clique, give the vertices at distance 1, 2 and 3; no distance exceeds 3.

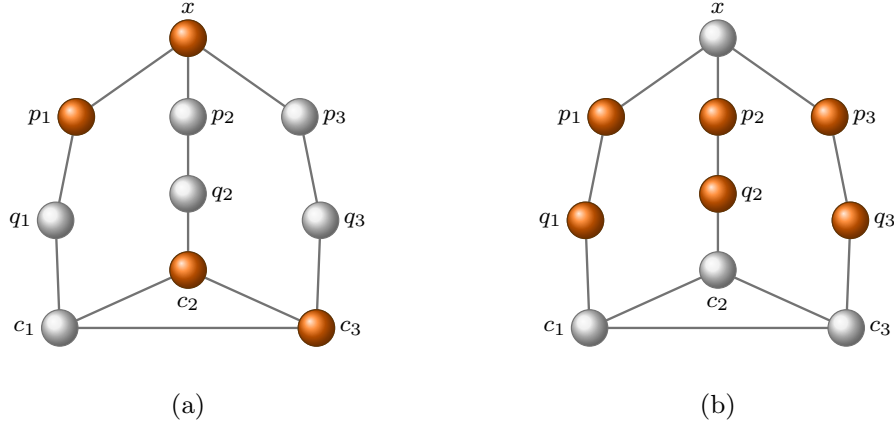


FIGURE 3. G_3 , the smallest counterexample to Conjecture 319. Every vertex has four vertices at even distance (itself included) and $\gamma(G_3) = 4$. The shaded sets are minimal total dominating sets of sizes 4 (a) and 6 (b).

vertex	distance 1	distance 2	distance 3	de
x	p_i	q_i	c_i	$1 + k$
p_i	x, q_i	p_j, c_i	q_j, c_j	$1 + (k - 1) + 1$
q_i	p_i, c_i	x, c_j	p_j, q_j	$1 + 1 + (k - 1)$
c_i	c_j, q_i	p_i, q_j	x, p_j	$1 + 1 + (k - 1)$

For instance $d(p_i, c_j) = 3$ along p_i, q_i, c_i, c_j , while the route through x has length 4; and $d(q_i, q_j) = 3$ along q_i, c_i, c_j, q_j . In every row $\text{de} = k + 1$.

Domination. The k sets $N[q_i] = \{p_i, q_i, c_i\}$ are disjoint, so a dominating set D has at least k vertices. Suppose $|D| = k$. Then D has exactly one vertex in each $N[q_i]$ and does not contain x , so x is dominated by some $p_i \in D$. Then $c_i \notin D$ and $q_i \notin D$, so c_i is dominated by some $c_j \in D$, $j \neq i$. But then $p_j, q_j \notin D$ and $x \notin D$, so p_j is not dominated, a contradiction. Hence $\gamma(G_k) \geq k + 1$.

The set $A = \{x, p_1, c_2, \dots, c_k\}$ is a total dominating set: x is dominated by p_1 ; every p_i by x ; q_1 by p_1 ; q_i by c_i for $i \geq 2$; c_1 by c_2 ; and c_i , $i \geq 2$, by another c_j with $j \geq 2$, which exists since $k \geq 3$. As $\gamma \leq \gamma_t \leq |A| = k + 1$, we get $\gamma(G_k) = \gamma_t(G_k) = k + 1$, and A is minimal because it has the least possible size.

The set $B = \{p_1, \dots, p_k, q_1, \dots, q_k\}$ is a total dominating set: x is dominated by p_1, p_i and q_i by each other, and c_i by q_i . It is minimal: without p_i the vertex q_i has no neighbour in B , since $N(q_i) = \{p_i, c_i\}$; without q_i the vertex c_i has none, since $N(c_i) = \{q_i\} \cup \{c_j : j \neq i\}$. As $2k > k + 1$, G_k is not well totally dominated, while $\max_v \text{de}(v) = k + 1 = \gamma(G_k)$. $\square$

Remark 5.3. If v itself is not counted in $\text{dist}_{\text{even}}(v)$, Conjecture 319 already fails for the path $v_1 v_2 v_3 v_4 v_5$: the largest count is $2 = \gamma(P_5)$, while $\{v_2, v_3, v_4\}$ and $\{v_1, v_2, v_4, v_5\}$ are minimal total dominating sets. So the conjecture is false under either reading; the definition in [5] counts all vertices at even distance, which includes v .

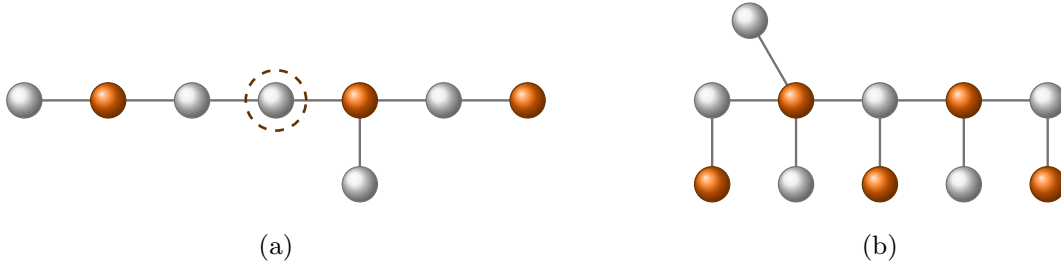


FIGURE 4. (a) The smallest counterexample to Conjecture 427: the right side is 2 (the center is circled) and $i = 3$, attained by the shaded set. (b) F_5 : with C read as the set of maximum-degree vertices the right side is 4, while $i = 5$.

6. CONJECTURE 427

Definition 6.1. For $m \geq 2$ let H_m be the path $v_{-m}, \dots, v_0, \dots, v_m$ with one leaf ℓ_j attached to each v_j , $j \neq 0$; it has $4m + 1$ vertices. For $m \geq 3$ let F_m be the path $v_1, \dots, v_m$ with one leaf attached to each v_j and a second leaf attached to v_2 ; it has $2m + 1$ vertices.

Theorem 6.2. For $m \geq 2$, the right side of Conjecture 427 for H_m equals 2, while $i(H_m) = 2m$. If C is replaced by the set M of maximum-degree vertices, the right side for F_m equals 4, while $i(F_m) = m$; so for $m \geq 5$ the conjecture fails in this reading too.

Proof. In H_m the path $\ell_{-m}, v_{-m}, \dots, v_m, \ell_m$ is a longest path, so $C = \{v_0\}$ by Jordan's theorem. As v_0 has degree 2, $|E(C, V - C)| = 2$. The pendant vertices are the ℓ_j , so $N(P) = \{v_j : j \neq 0\}$ and $V - N(P) = \{v_0\} \cup \{\ell_j\}$, an independent set; the second term vanishes and the right side is 2. By Lemma 2.1 an independent dominating set meets each of the $2m$ disjoint sets $\{v_j, \ell_j\}$, $j \neq 0$, so $i(H_m) \geq 2m$; and $\{v_j : j \text{ odd}\} \cup \{\ell_j : j \neq 0 \text{ even}\}$ is an independent dominating set of size $2m$ (it dominates v_0 through $v_{\pm 1}$).

In F_m the vertex v_2 has degree 4 and every other vertex has degree at most 3, so $M = \{v_2\}$ and $|E(M, V - M)| = 4$. Every spine vertex is a support vertex, so $V - N(P)$ is the set of leaves, which is independent; the right side is 4. By Lemma 2.1, $i(F_m) \geq m$, and the vertices v_j with j even together with the leaves at the v_j with j odd form an independent dominating set of size m . $\square$

The smallest counterexample has 8 vertices: the path $a_1 a_2 \dots a_7$ with a leaf b attached to a_5 (Figure 4(a)). The path $a_1 \dots a_7$ is a longest path, so the center is $\{a_4\}$, which has degree 2. The pendant vertices are a_1 , a_7 and b , so $N(P) = \{a_2, a_5, a_6\}$, and $\langle V - N(P) \rangle$ has the single edge $a_3 a_4$. The right side is $2 + \lfloor \frac{2}{3} \rfloor = 2$. On the other hand $N[a_5]$ has four vertices and every other closed neighbourhood at most three, so two vertices dominate at most seven of the eight vertices; and $\{a_2, a_5, a_7\}$ is an independent dominating set. So $i = 3$.

7. EXHAUSTIVE SEARCH

The trees with at most 24 vertices (63,242,254 trees with 3 to 24 vertices) were generated with **gentreeg** from nauty [12], which implements the algorithm of Wright, Richmond, Odlyzko and McKay [13], and tested against Conjectures 352, 358 and 359; γ_t was computed by a dynamic programme over a rooted tree, and compared with a brute-force computation for all trees with at most 16 vertices. Table 1 gives the numbers of counterexamples. The connected graphs were generated with **geng**. Among the 1,018,690,328 connected graphs with 2 to 11 vertices, 46

TABLE 1. Counterexamples among all trees with n vertices.

n	trees	Conjecture 352	Conjectures 358, 359
3–17	81,135	0	0
18	123,867	1	0
19	317,955	4	1
20	823,065	13	3
21	2,144,505	31	14
22	5,623,756	73	38
23	14,828,074	157	108
24	39,299,897	318	255

satisfy the hypothesis of Conjecture 319, and G_3 is the only one of them that is not well totally dominated. Among the 11,989,760 connected graphs with 4 to 10 vertices, Conjecture 427 has 1, 12 and 72 counterexamples with 8, 9 and 10 vertices and none smaller; with C read as M it has 5, all with 10 vertices.

APPENDIX A. VERIFICATION

The proofs above use no computation. The computer checks the finite claims about the smallest members of each family and the searches of Section 7, independently in four languages; all files are in the repository listed under Data and code availability. The author used Claude (Anthropic), an AI system, for programming, computation and drafting; the author checked the mathematics and is responsible for the content.

- (1) **C.** Exhaustive searches: `search352.c` and `search358.c` over trees in graph6 format (total domination number by dynamic programming, optionally compared with brute force over all vertex subsets), `c319.c` and `c427.c` over connected graphs (all parameters by exhaustion over vertex subsets, in integer arithmetic). The search for Conjecture 319 on the 1,006,700,565 connected graphs with 11 vertices was split into three parts with `geng`'s `res/mod` option.
- (2) **Python** (exact integer and rational arithmetic): for $T(a, r)$ with $a \in \{3, 7, 11\}$, $r \in \{4, 8, 12, 16\}$, for Q_k with odd $k \leq 11$, for G_k with $k \leq 5$, for H_m with $m \leq 5$ and for F_m with $5 \leq m \leq 8$, recomputes every quantity in Theorems 3.2–6.2 from the edge lists, and checks the stated values of γ_t , γ and i by exhaustion for the smaller members.
- (3) **Julia**: the same quantities, with γ_t , γ , i and the minimal total dominating sets found by brute force over all vertex subsets (up to 2^{26} subsets), except that $\gamma_t(Q_k)$ for $k \geq 5$ is computed by a tree dynamic programme.
- (4) **Lean 4**: self-contained files without Mathlib. For $T(3, 4)$, Q_3 , G_3 , the 8-vertex tree and F_5 they compute the right sides of the conjectures from the edge lists and prove that the conjectures fail. The refutations of Conjectures 352, 358 and 359 exhibit a total dominating set that is too small and depend on no axioms; those of Conjectures 319 and 427 evaluate γ and i over all vertex subsets in the kernel and depend only on `propext`. The exact values $\gamma_t(T(3, 4)) = 7$ and $\gamma_t(Q_3) = 9$ are also checked, by compiled evaluation (`native_decide`); the refutations do not use them.

STATEMENTS AND DECLARATIONS

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Data and code availability. The programs, their outputs and the Lean files are available at <https://github.com/creelie/eCY>.

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